

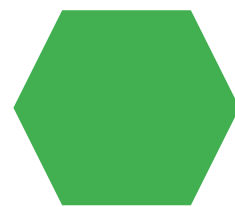


Seasonal Agriculture Performance Analysis

ABHISHEK M

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PROBLEM STATEMENT

- (Agricultural activities are influenced by seasonal variations in environmental conditions, farming practices, resource availability and market conditions. As a result, agricultural performance may differ from one season to another. However, raw agricultural data does not clearly explain how agricultural performance changes across seasons or what patterns can be observed in different seasonal conditions. The problem is to analyze the given agricultural dataset and investigate seasonal differences in agricultural performance by identifying meaningful patterns, trends, relationships and variations within the available data.



Seasonal variations can influence different aspects of agricultural activities and performance.

Understanding seasonal patterns through data analytics can help stakeholders:



- **Understand variations in agricultural performance**
- **Identify important seasonal trends**
- **Compare performance across different periods**

Understand changing environmental conditions

Examine differences in resource usage

- **Identify areas requiring further investigation**
- **Support evidence-based agricultural planning**



WHO ARE THE END USERS?



Farmers & Producer Organizations (FPOs):
Agricultural Policy Makers & Planners
Irrigation & Water Resource Authorities:
Agri-Financiers & Insurance Underwriters

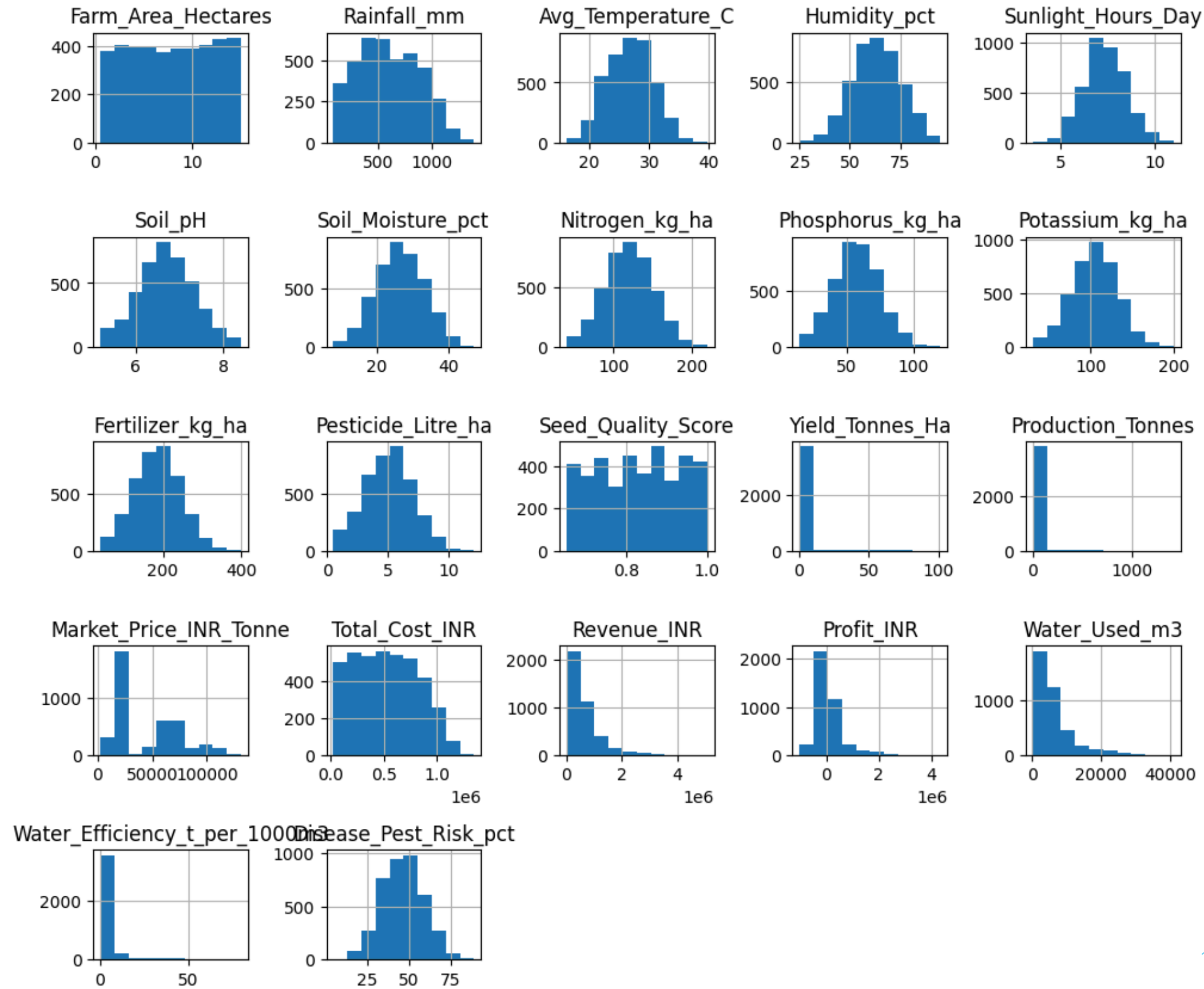


Technology Used

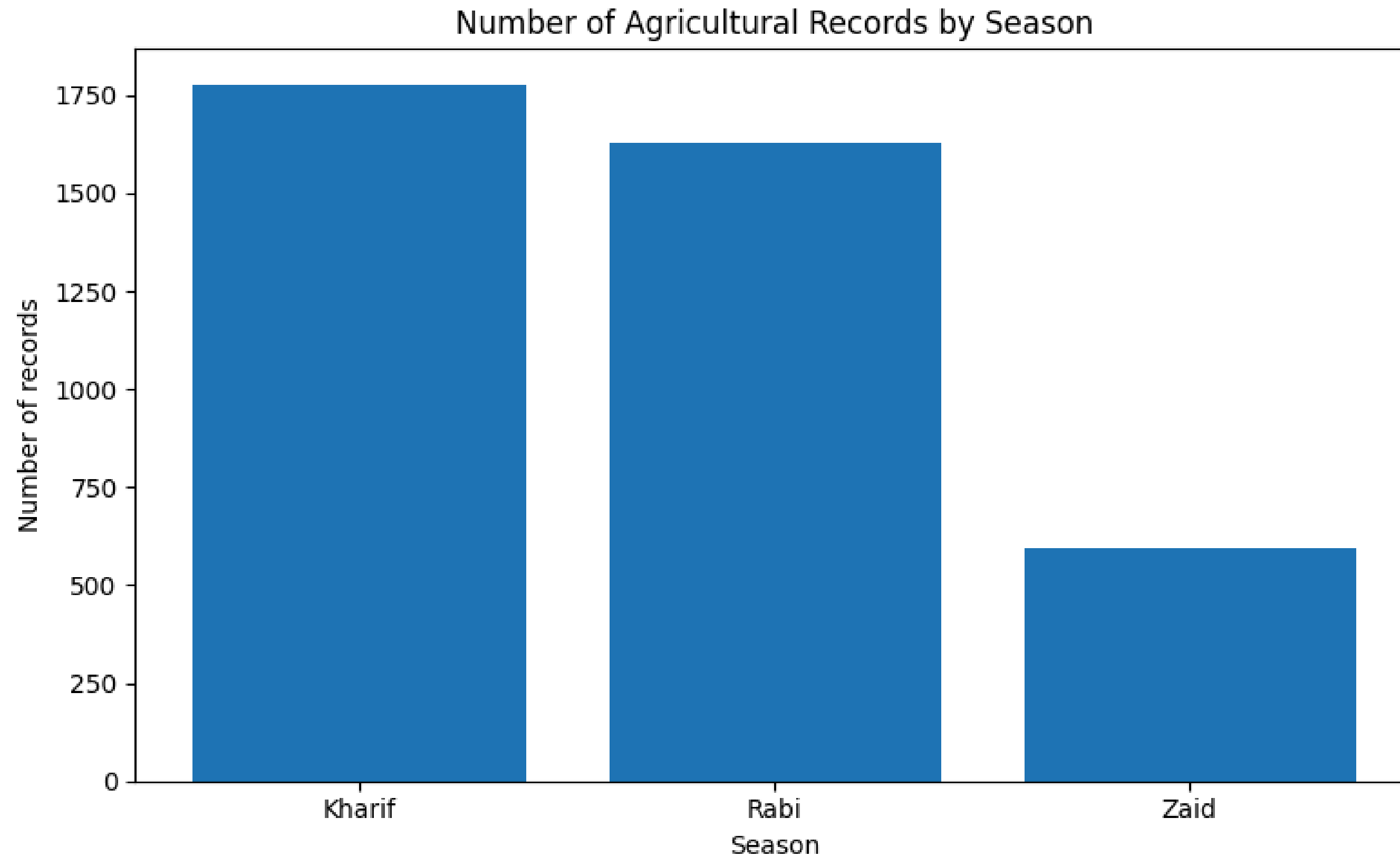
- Programming & Execution: Python 3.x, Jupyter Notebook
- Data Engineering & Manipulation: Pandas, NumPy
- Statistical Analysis & Visualization: Matplotlib, Seaborn
- Development Environment & Version Control: Git & GitHub



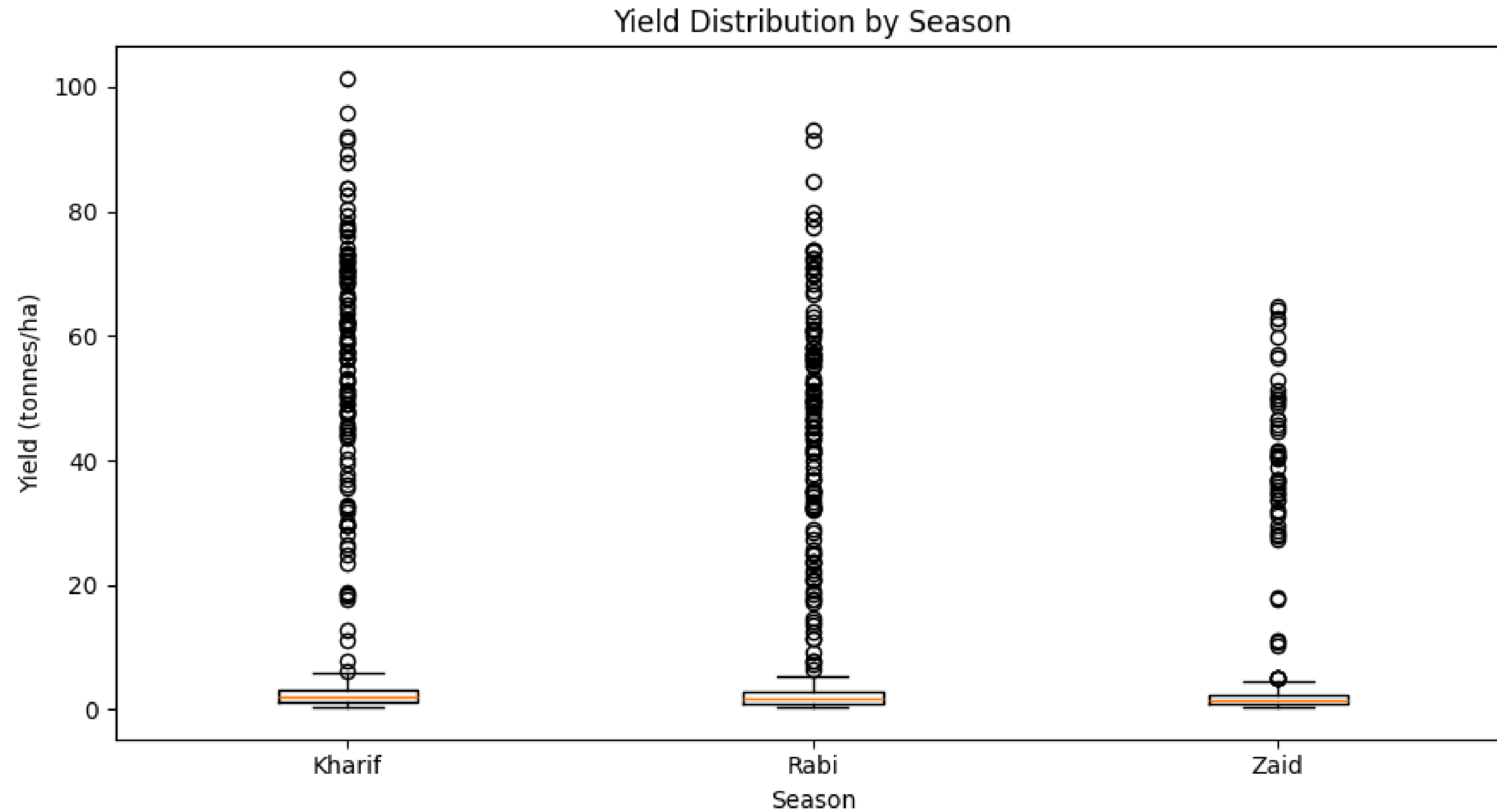
RESULTS



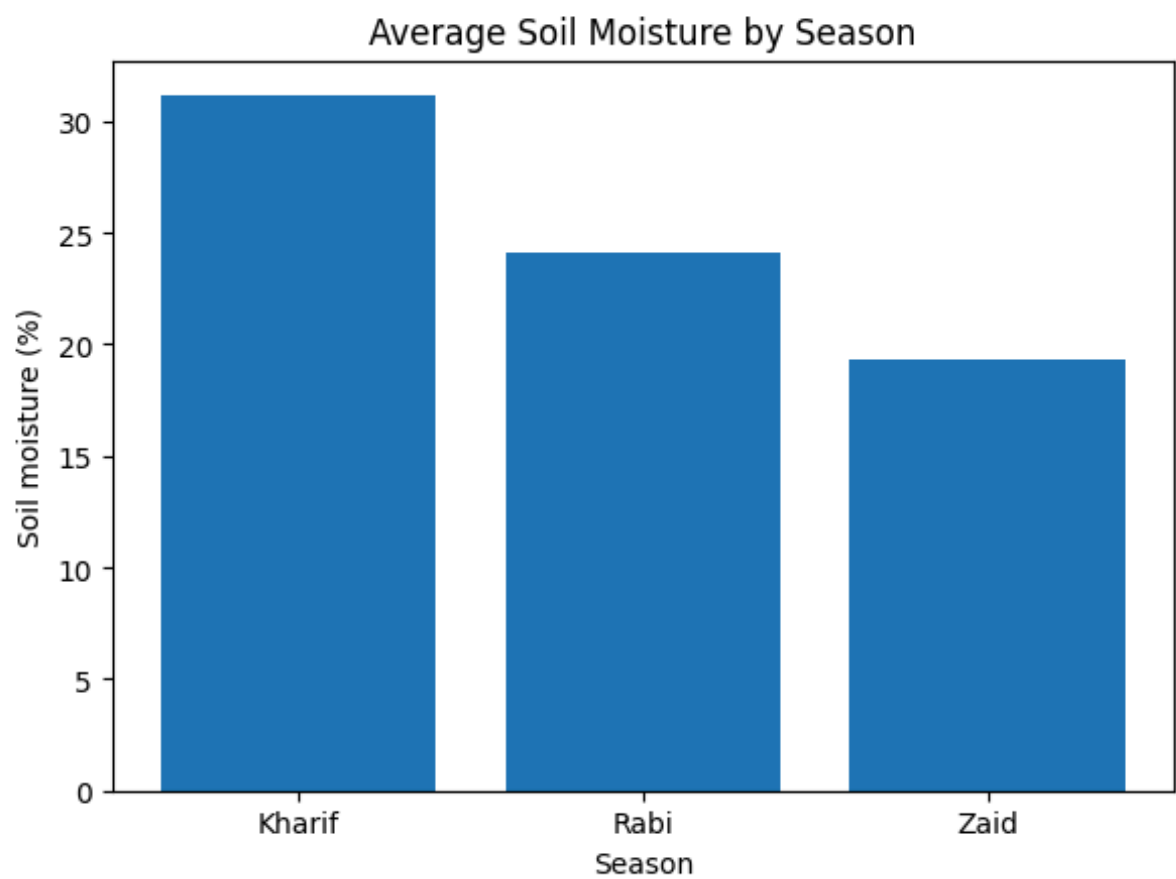
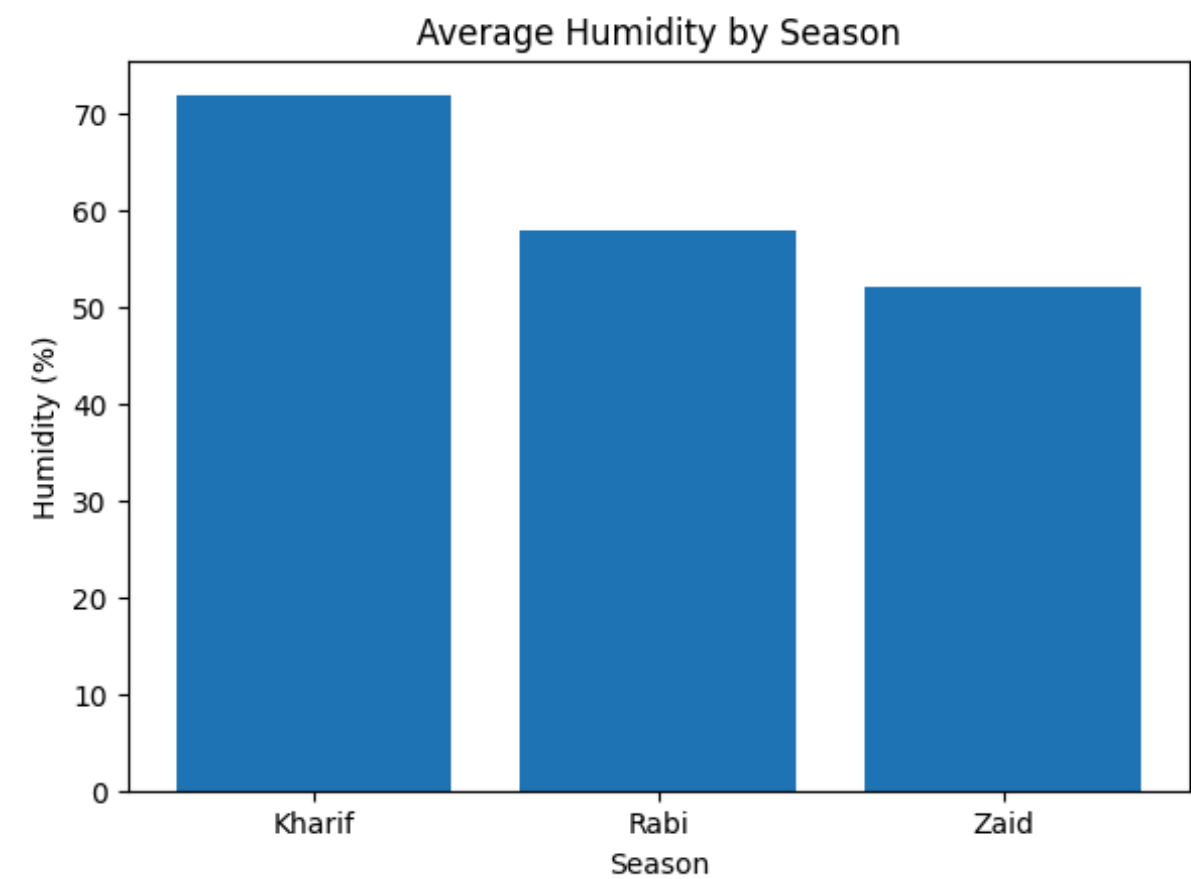
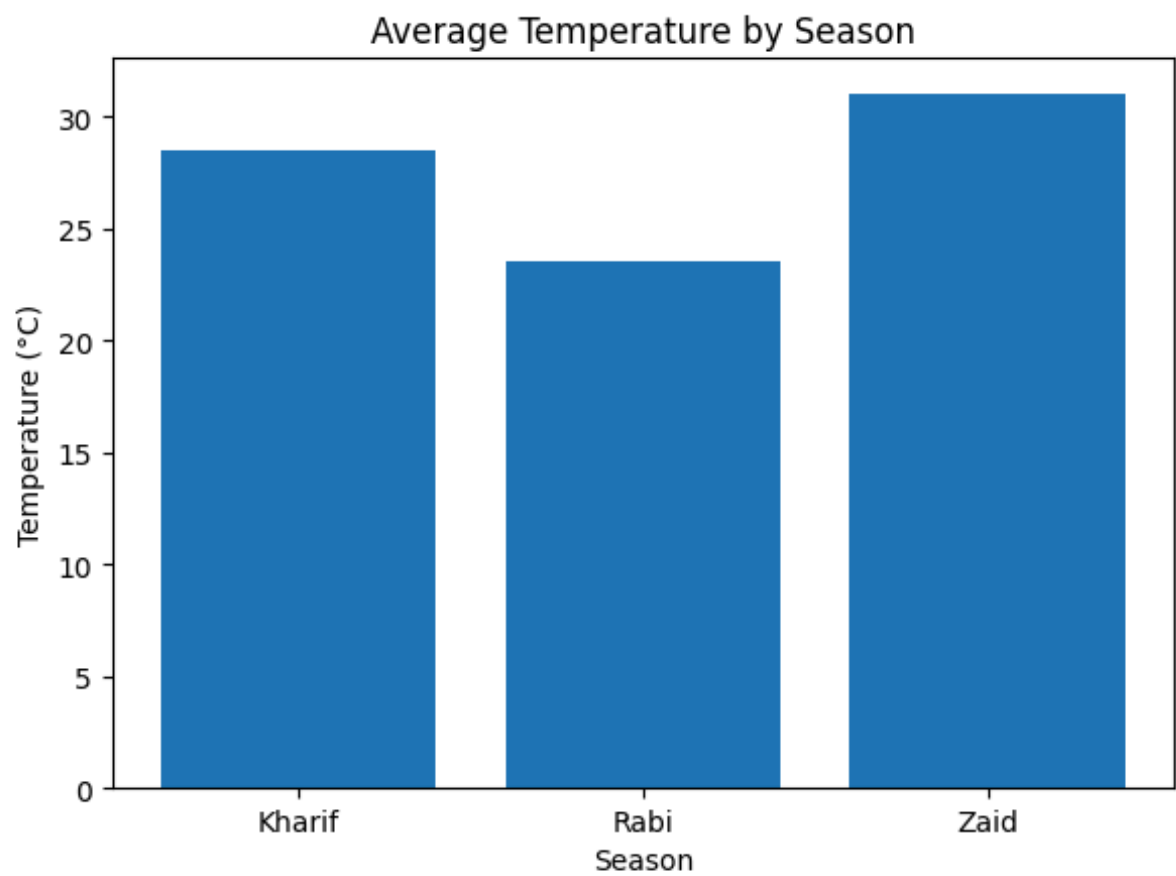
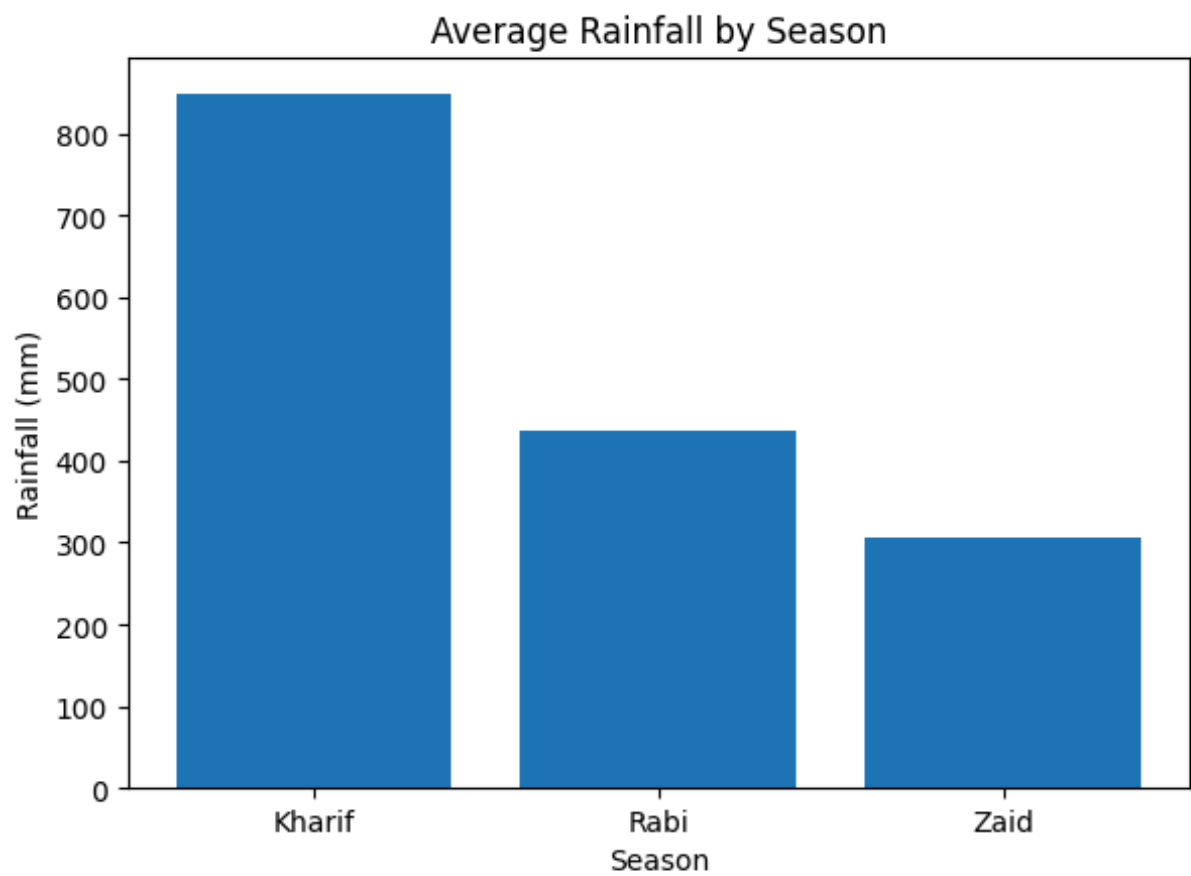
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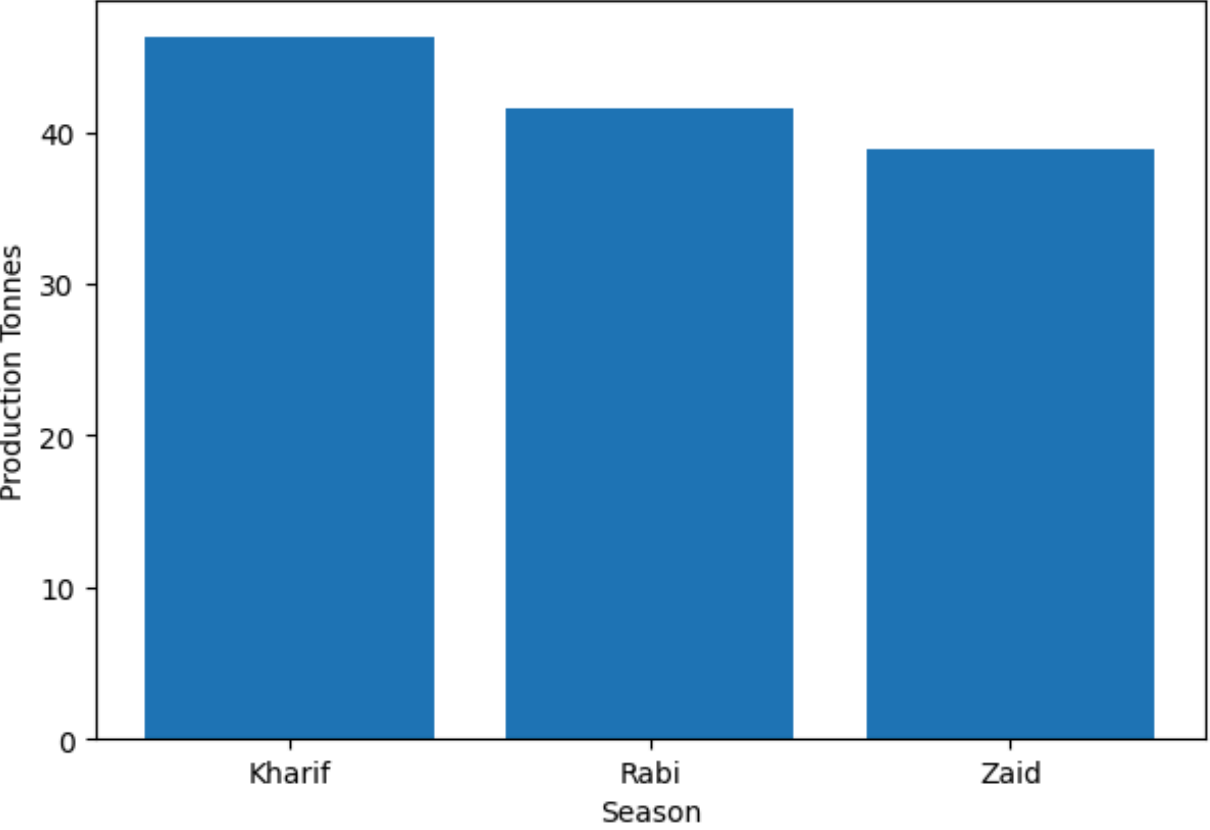


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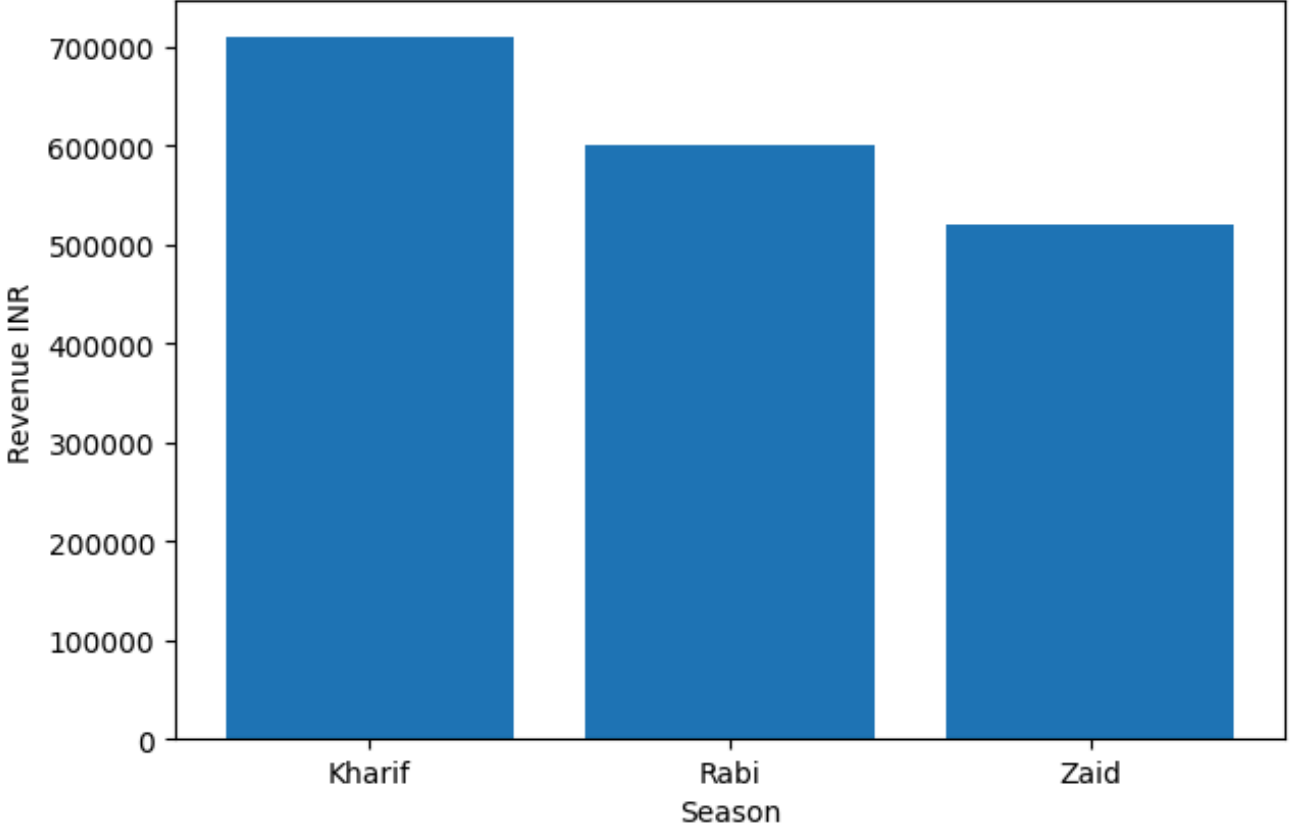


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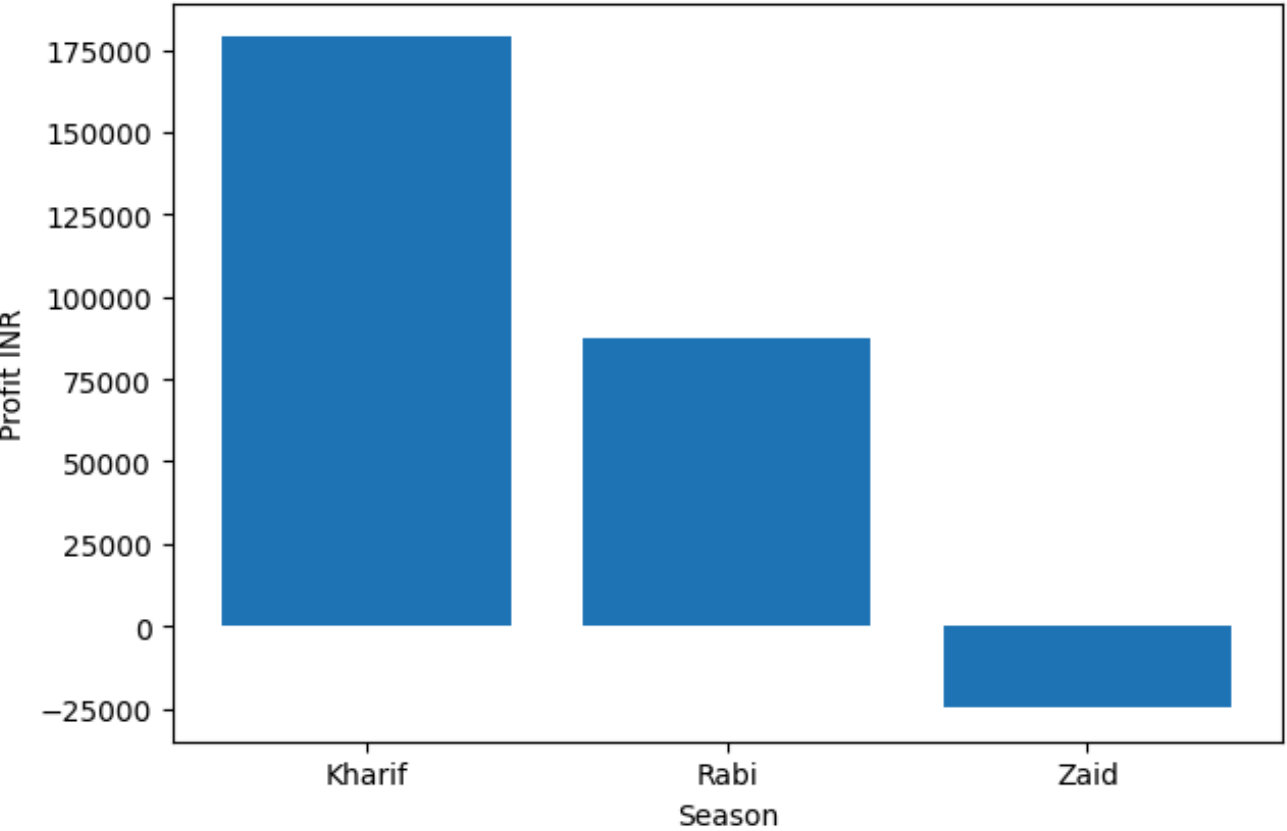
Average Production by Season



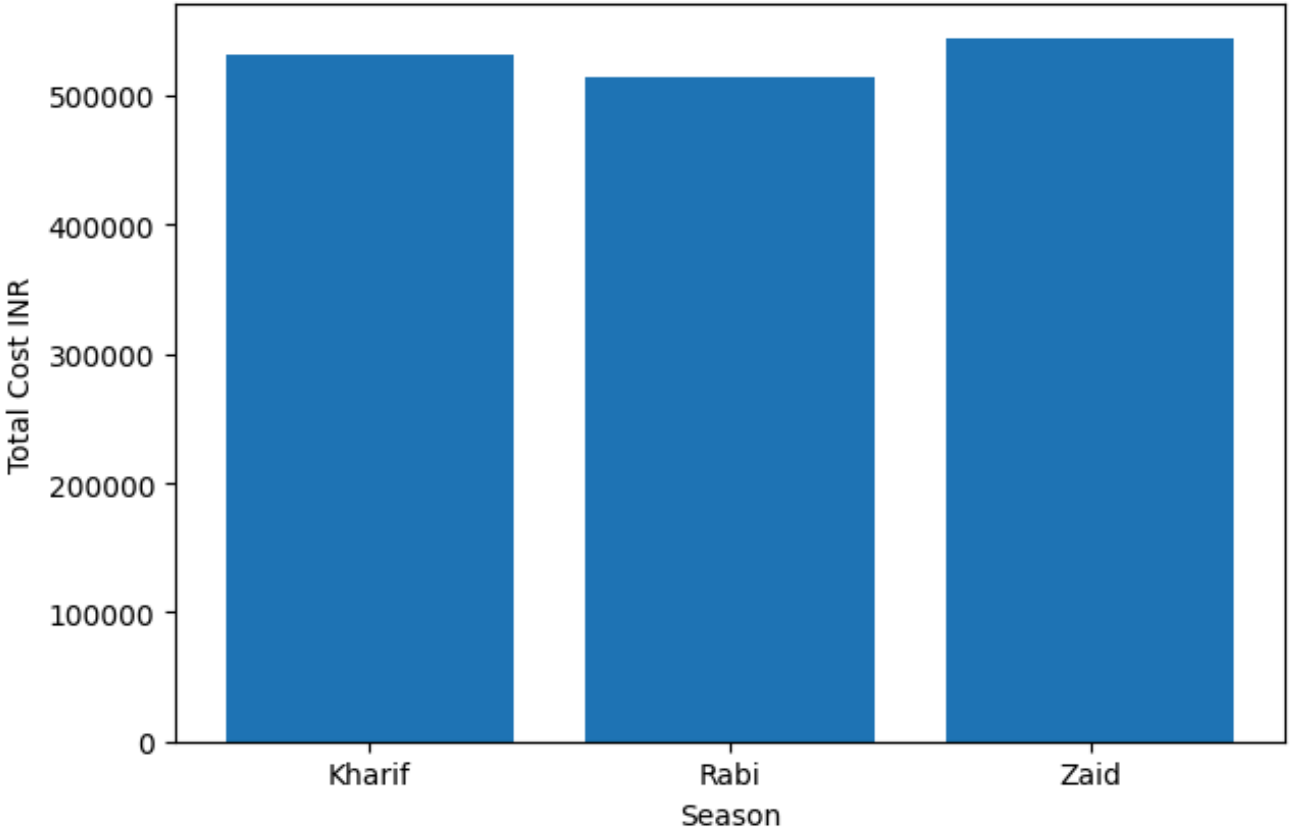
Average Revenue by Season



Average Profit by Season

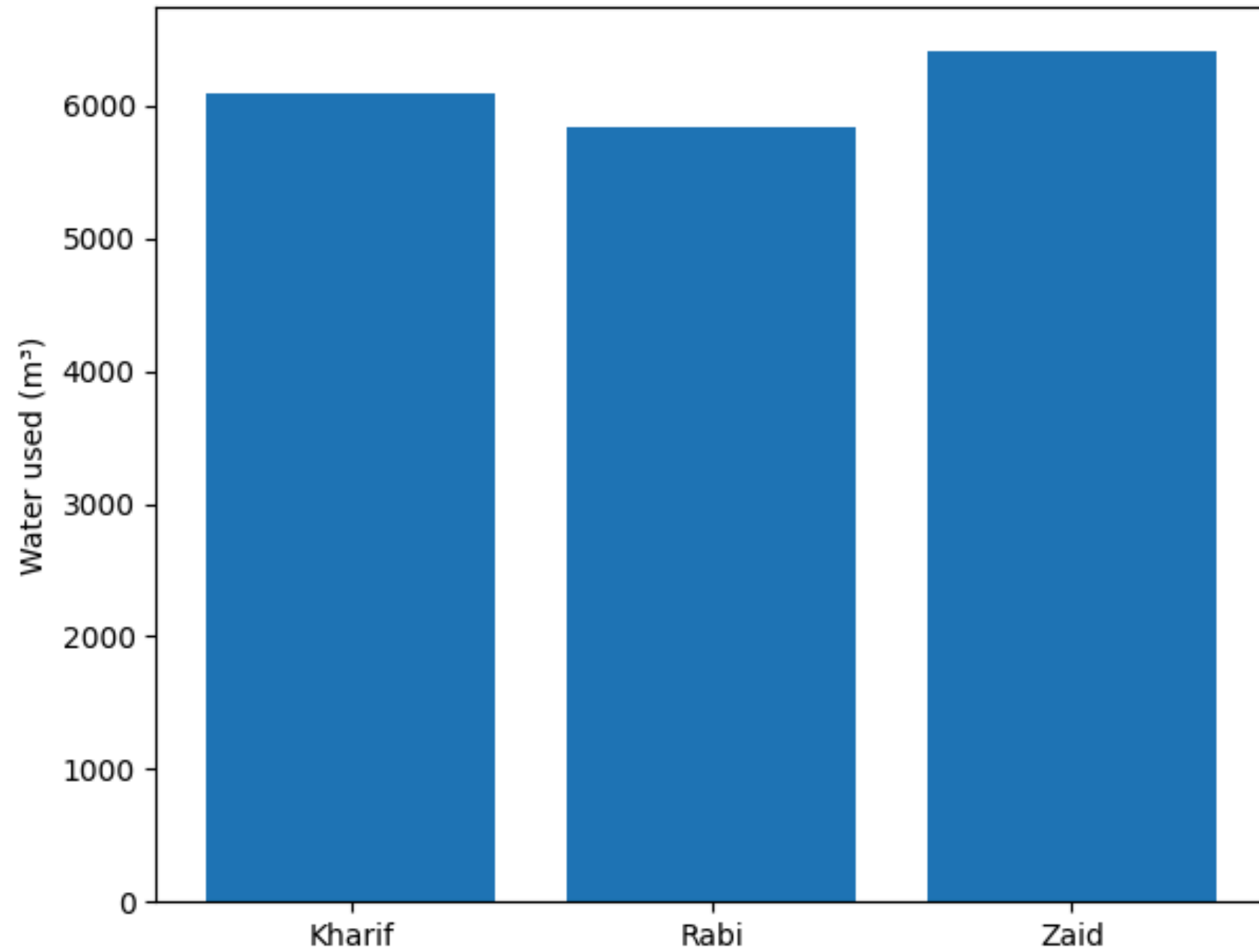


Average Total Cost by Season

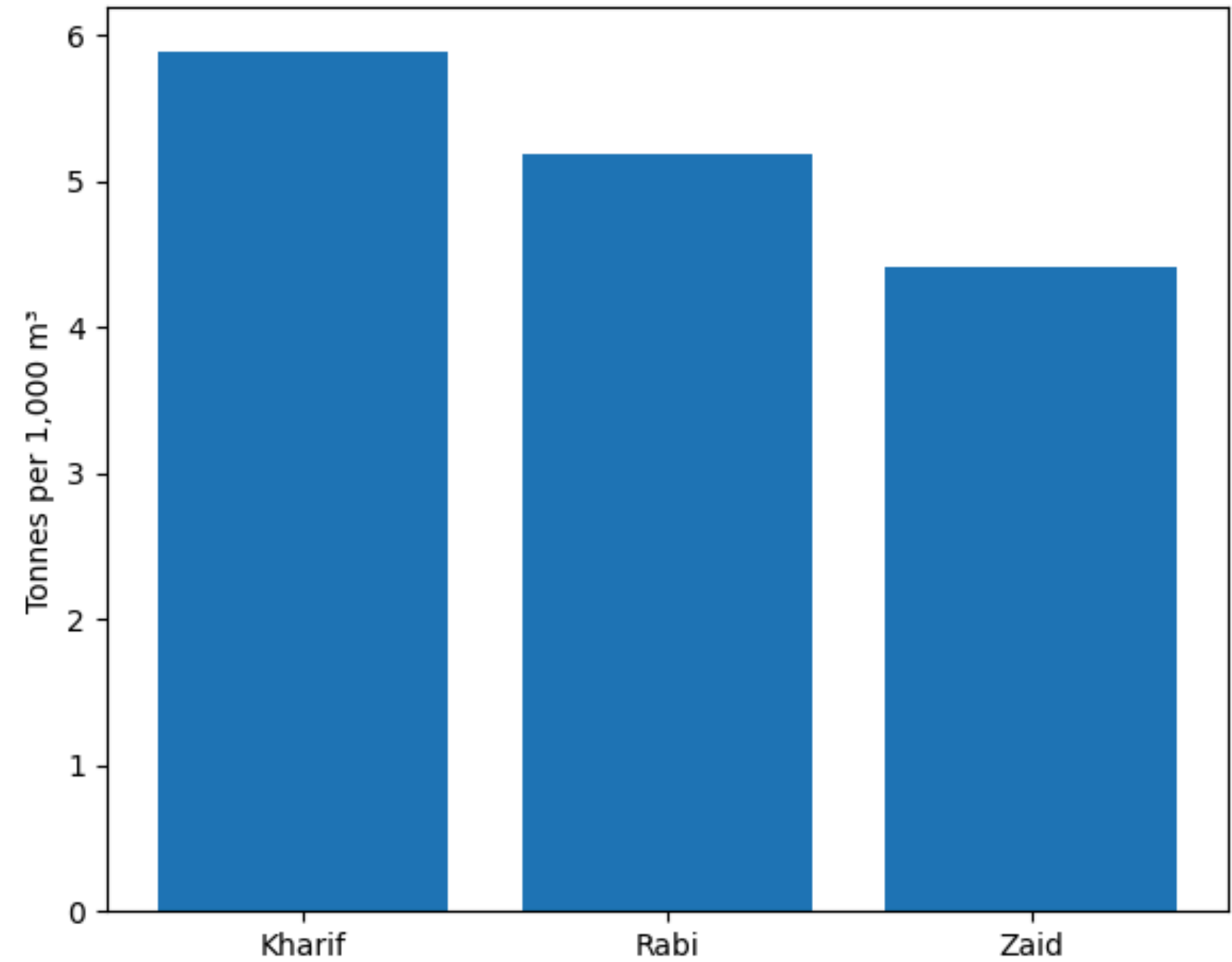


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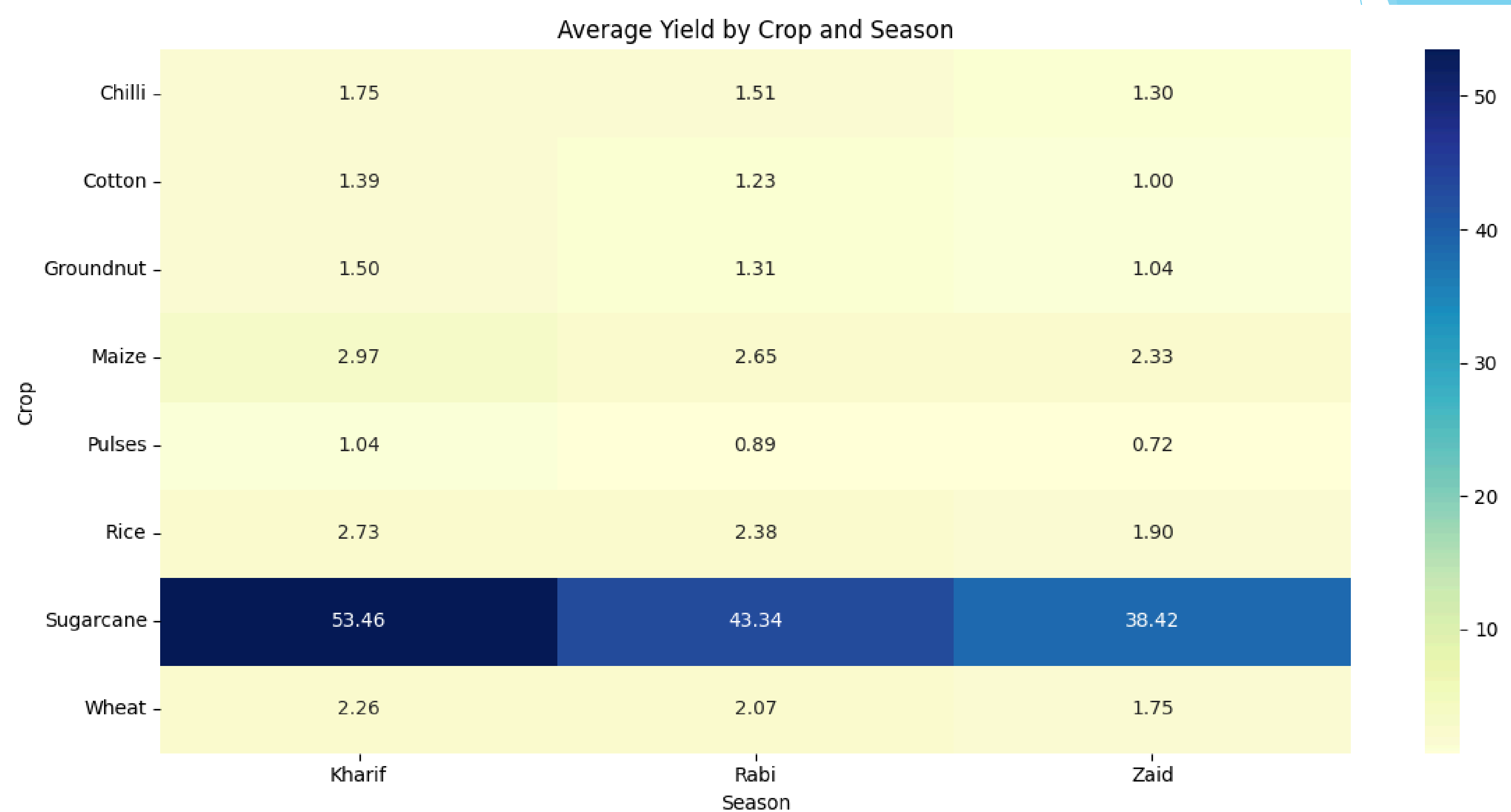
Average Water Used by Season



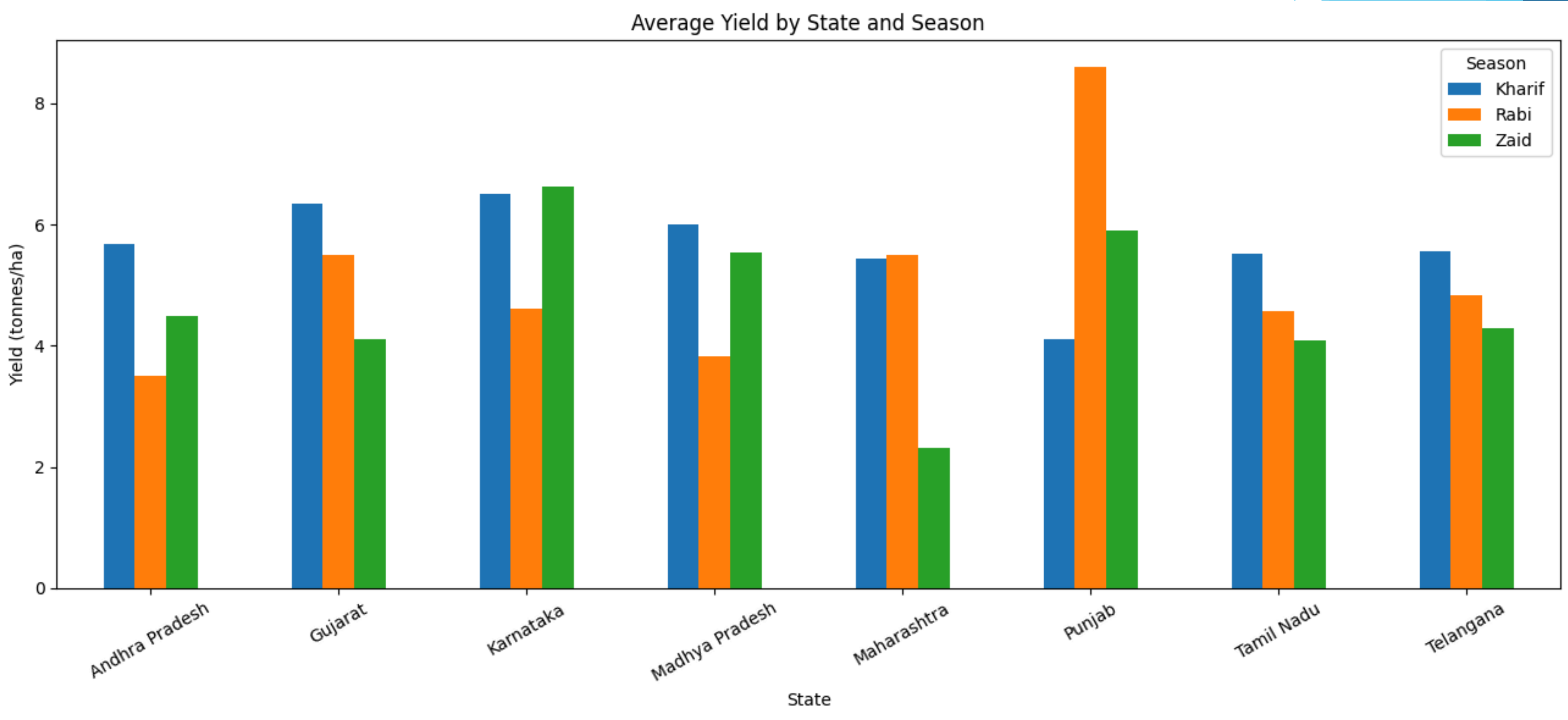
Average Water Efficiency by Season



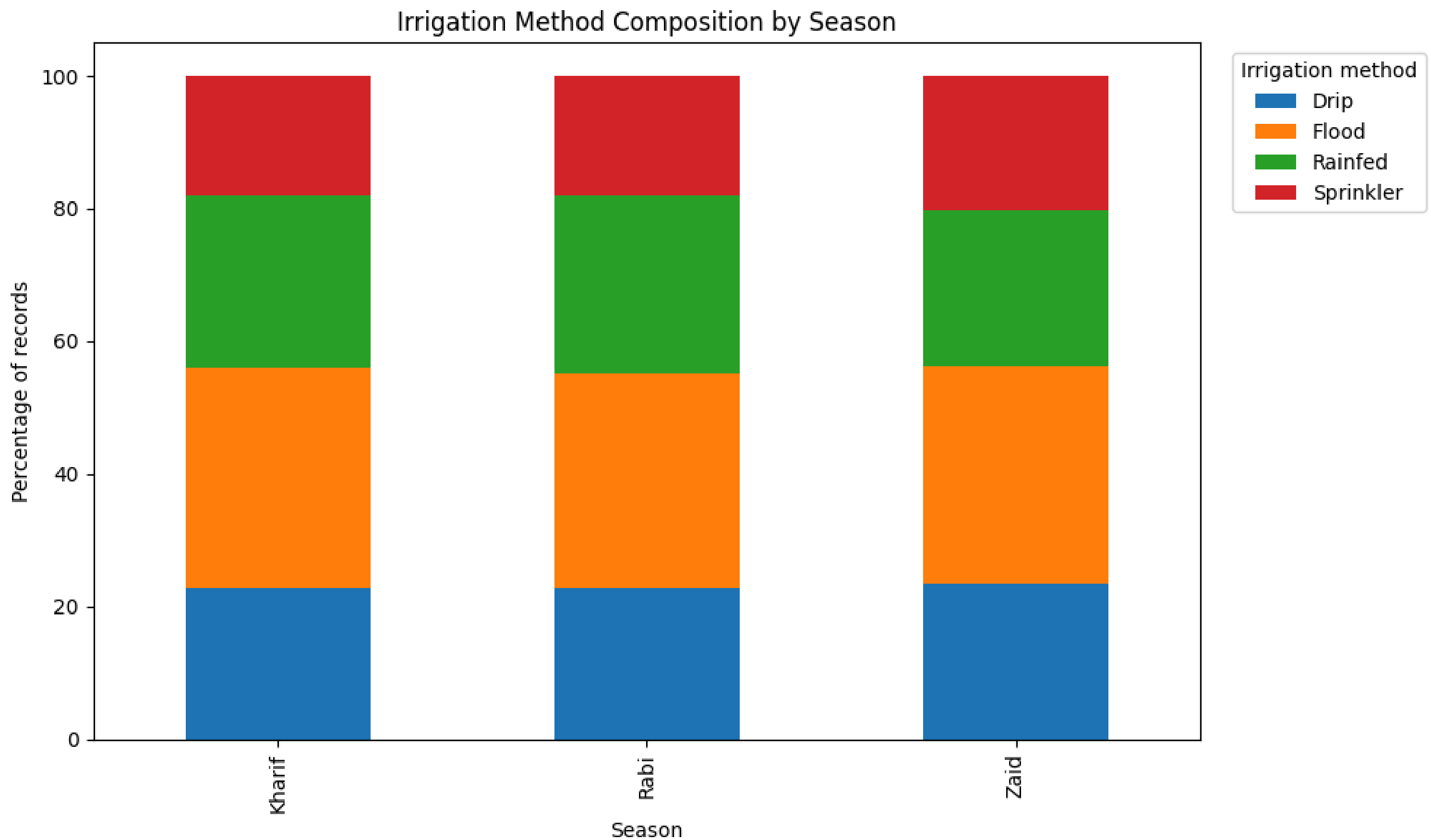
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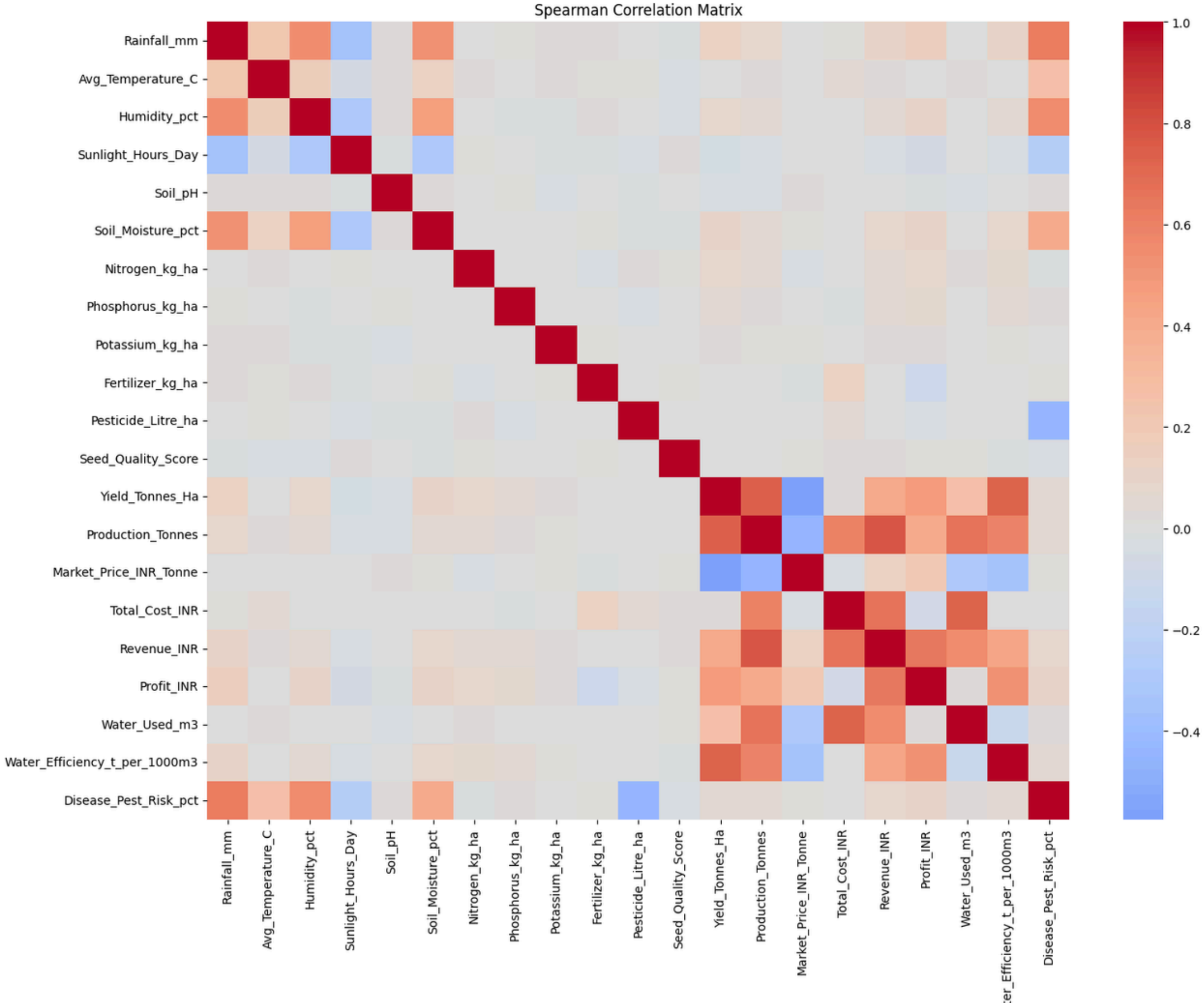
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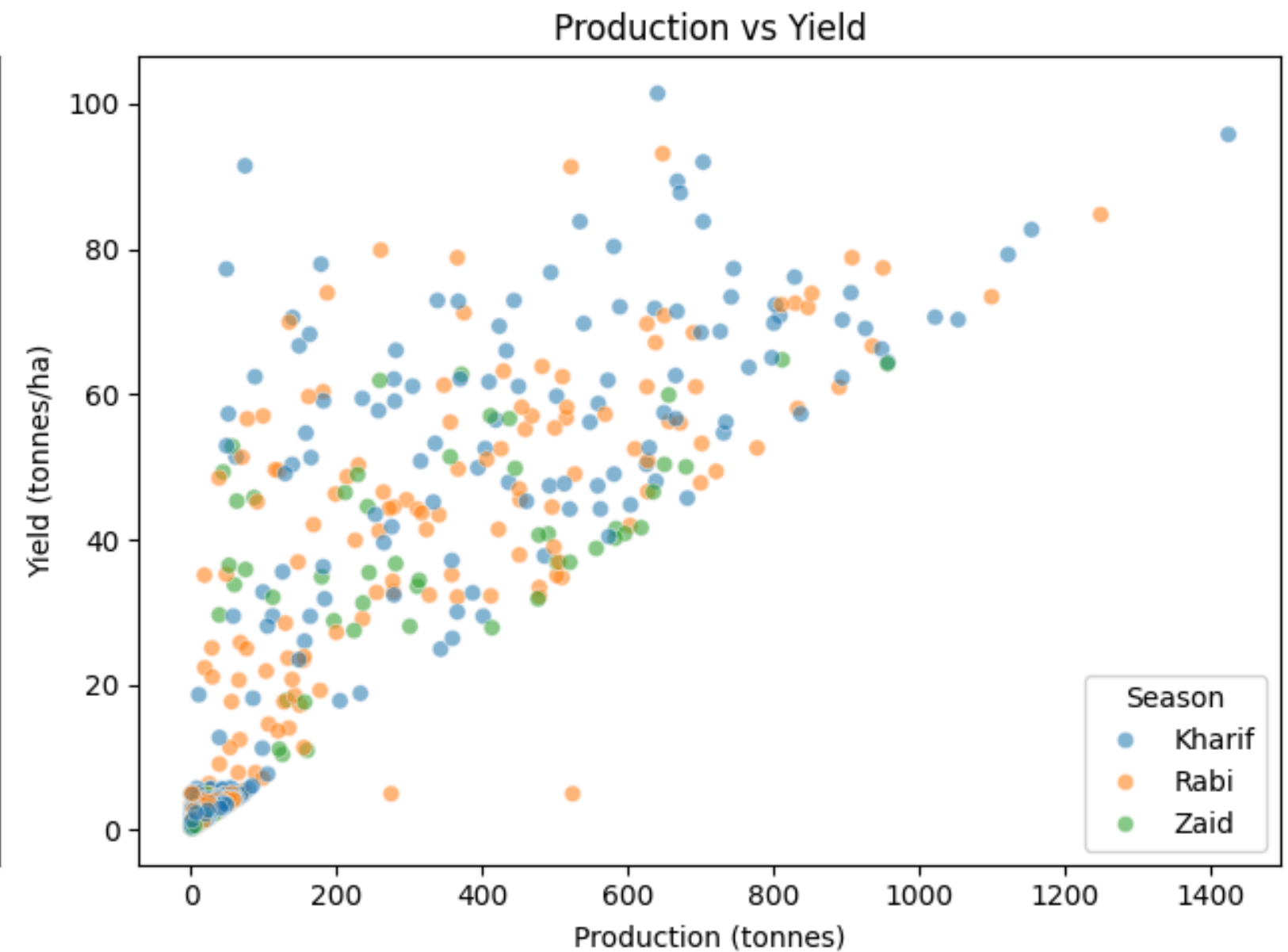
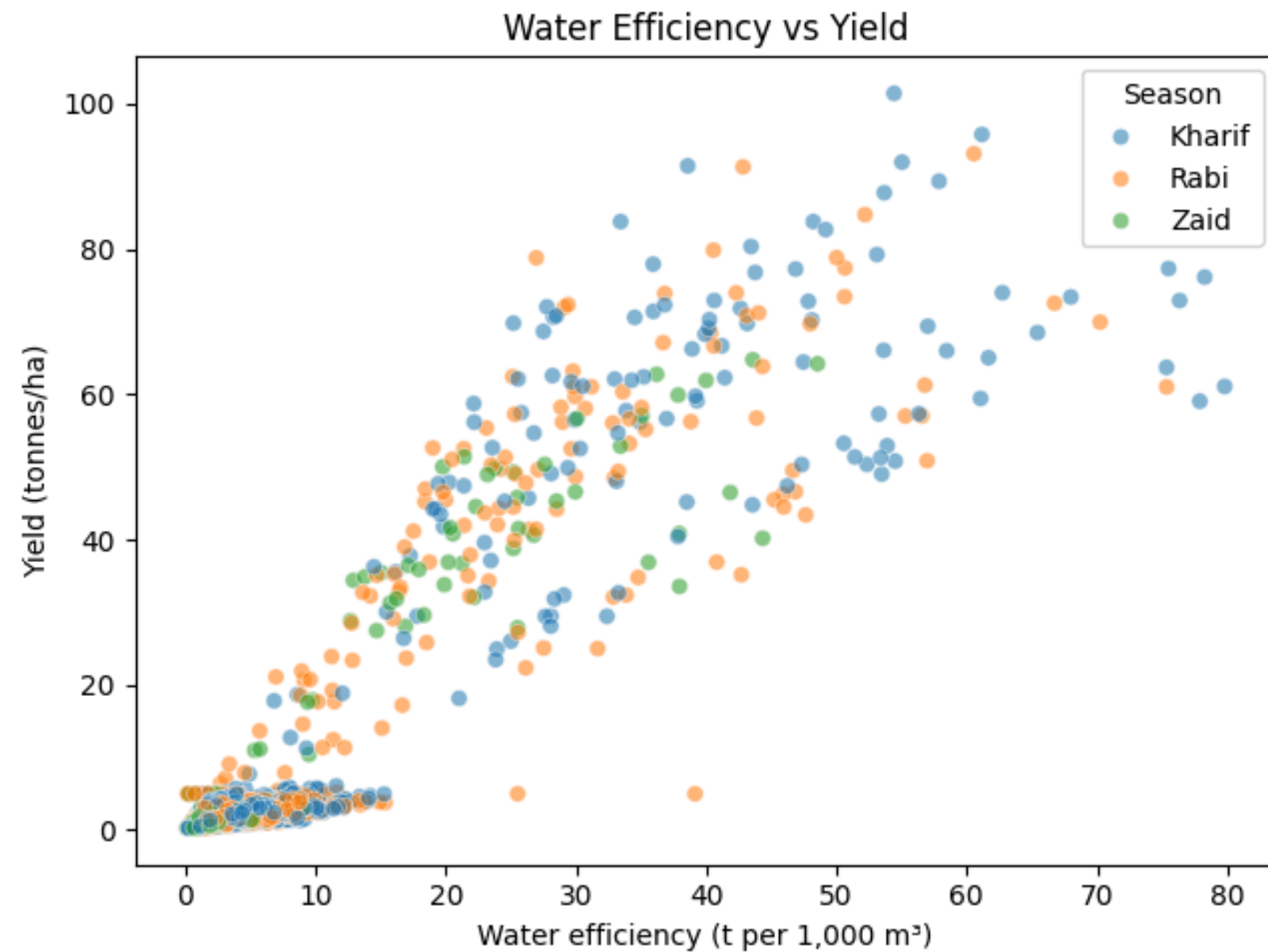
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Future scope

Predictive Machine Learning: Implement gradient boosting models (e.g., XGBoost, LightGBM) to forecast crop yields and optimal harvest windows prior to sowing.

Prescriptive Resource Allocation: Build an automated decision support engine that matches crop selection to real-time ground moisture and local market price forecasts.

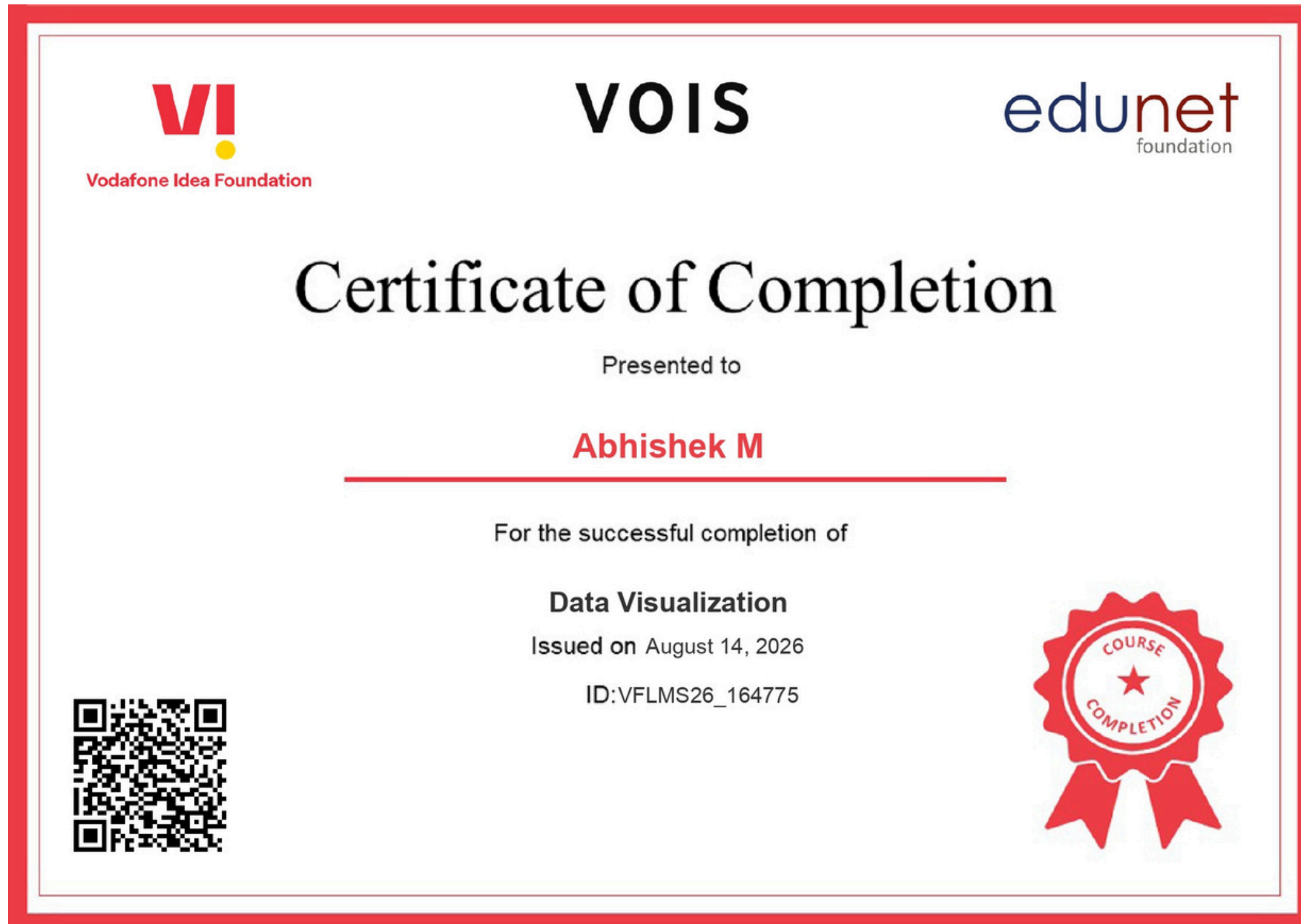
Integration with Remote Sensing: Incorporate satellite-derived NDVI indices and IoT weather stations to detect microclimatic anomalies before yield degradation occurs.

GitHub Link

<https://github.com/AswiniKumar55/-Seasonal-Agriculture-Performance-Analysis-.git>



VOIS Course completion certificate - Course name- Data Visualization



THANK YOU

